

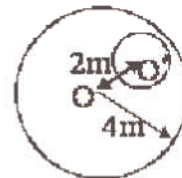
**WORK SHEET - 08****ROTATION DYNAMICS**

- Q.1 A rope of mass 'm' is looped in a circle of radius R and rotated with a constant angular velocity ω_0 about its axis in gravity free space. Find the tension in the rope ?

(A) $T = mR\omega_0^2$ (B) $2mR\pi\omega^2$ (C) $T = \frac{mR\omega_0^2}{2\pi}$ (D) $4mR\pi\omega_0^2$

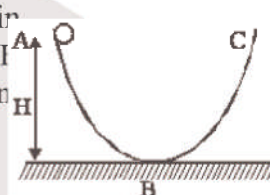
- Q.2 From a disc of mass 2 kg and radius 4m a small disc of radius 1 m with center is extracted. The new moment of inertia, about an axis passing through O perpendicular to plane of disc is

(A) 16 kg m^2 (B) 12 kg m^2 (C) $\frac{255}{16} \text{ kgm}^2$ (D) $\frac{247}{16} \text{ kgm}^2$



- Q.3 A sphere of mass m and radius r is released from rest at point A on a track in vertical plane. The track is rough enough to support rolling between A and I and from B onwards it is smooth. The maximum height attained by sphere from ground on its journey from B onwards is

(A) H (B) $\frac{5}{7}H$ (C) $\frac{2}{5}H$ (D) $\frac{2}{7}H$



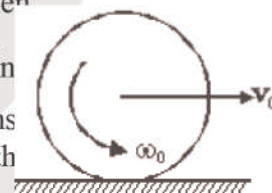
- Q.4 Two particles A and B of mass m each and moving with velocity v, hit the ends of a rigid bar of the same mass m and length l simultaneously and stick to the bar as shown in the figure. The bar is kept on a smooth horizontal plane. The linear and angular speed of the system (bar + particle) after the collision are

(A) $v_{cm} = 0, \omega = \frac{12v}{7l}$ (B) $v_{cm} = 0, \omega = \frac{4v}{l}$
 (C) $v_{cm} = 0, \omega = \frac{5v}{l}$ (D) $v_{cm} = 0, \omega = \frac{v}{5l}$



- Q.5 A disc of radius R is spun to an angular speed ω_0 about its axis and then imparted a horizontal velocity of magnitude $\frac{\omega_0 R}{4}$ (at $t=0$) with its plane remaining vertical. The coefficient of friction between the disc and the plane is μ . The sense of rotation and direction of its linear speed are shown in the figure. Choose the correct statement. The disc will return to its initial position

(A) if the value of $\mu < 0.5$. (B) irrespective of the value of μ ($\mu > 0$).
 (C) if the value of $0.5 < \mu < 1$. (D) if $\mu > 1$.



- Q.6 A uniform rod of mass m and length ℓ starts rotating with constant angular acceleration α due to a constant couple in a horizontal plane about a fixed vertical axis passing through one end. The horizontal component of the net force exerted on the rod by the axis when it has rotated by an angle $\pi/2$, is

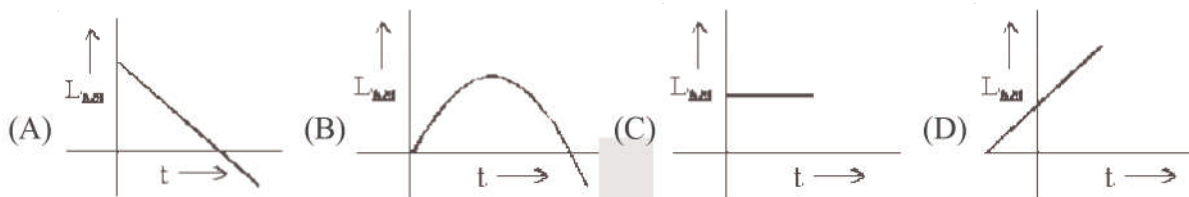
(A) $m\alpha\frac{\ell}{2}$ (B) $m\alpha\frac{\ell}{2}\sqrt{1+\pi^2}$ (C) $\frac{m\pi\alpha\ell}{2}$ (D) none of these

- Q.7 The point masses m_1 and m_2 are placed at the two ends of a rigid massless rod of length ℓ . The system is set into rotation with an angular velocity ω about an axis which is at a distance x from the mass m_1 and perpendicular to rod. The value of x for which the work done in the process is minimum is [3]

(A) $\frac{l}{2}$ (B) $\frac{m_1}{m_2}l$ (C) $\frac{m_1 l}{m_1 + m_2}$ (D) $\frac{m_2 l}{m_1 + m_2}$

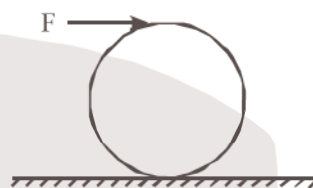


- Q.8 Two particles A and B starts moving simultaneously along two parallel lines separated by a distance ℓ in same direction. A starts moving with a constant velocity V_0 while B starts moving with a constant acceleration 'a' which of the following graph represents the angular momentum of A with respect to B in Vs time



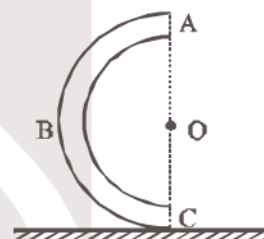
- Q.9 A ring of mass M is kept on a horizontal rough surface. A force F is applied tangentially at its rim as shown. The coefficient of friction between the ring and the surface is μ . Then

- (A) friction will act in the forward direction
(B) friction will act in the backward direction
(C) frictional force will not act
(D) frictional force will be μmg



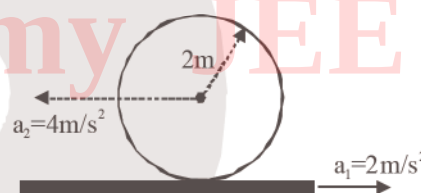
- Q.10 A half section of thin pipe of mass M and radius r is released from rest in the position shown. Pipe rolls without sliding. The change in potential energy of pipe after it has rolled through 90° .

- (A) $\frac{4mgr}{\pi}$ (B) $\frac{2mgr}{\pi}$
(C) mgr (D) $\sqrt{2} mgr$



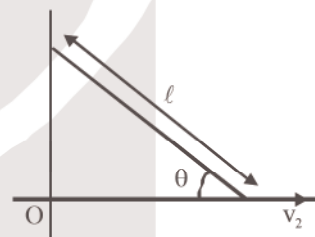
- Q.11 In the figure, a sphere of radius 2 metre rolls on a plank. The accelerations of the sphere and the plank are indicated. The value of α is

- (A) 2 rad/s^2 (B) 4 rad/s^2
(C) 3 rad/s^2 (D) 1 rad/s^2



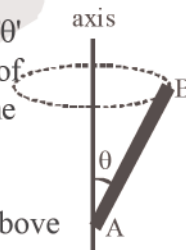
- Q.12 In the figure, there is a rigid rod of length ℓ which slides in such a way that it remains in contact with a fixed vertical wall and horizontal floor when the rod make angle θ with the horizontal, speed of lower end is v_2 . At this moment its angular speed is

- (A) $\frac{v_2}{\ell}$ (B) $\frac{v_2}{\ell \sin \theta}$ (C) $\frac{v_2 \sin \theta}{\ell}$ (D) $\frac{v_2}{\ell \cos \theta}$



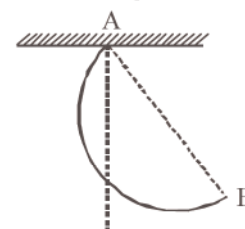
- Q.13 A uniform rod of mass 'm' and length 'L' is rigidly fixed to an axis, making an angle ' θ ' with it as shown in the figure. The rod is rotated about this axis so that the free end of the rod moves with a uniform speed 'v'. The angular momentum of the rod about the axis is

- (A) $\frac{1}{3} m v L \sin \theta$ (B) $\frac{1}{4} m v L \sin \theta$ (C) $m v L \cos \theta$ (D) none of the above



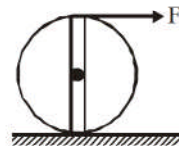
- Q.14 A uniform semicircular ring of mass m and radius r is hinged at end A so that it can rotate freely about end A in the vertical plane as shown in the figure. The angle made by the line AB with vertical in the equilibrium position is

- (A) 0 (B) $\tan^{-1}\left(\frac{2}{\pi}\right)$ (C) $\tan^{-1}\left(\frac{\pi-2}{\pi}\right)$ (D) $\tan^{-1}\left(\frac{\pi}{2}\right)$



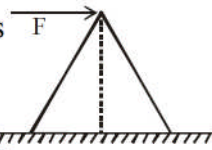
- Q.15 A circular ring of mass m and radius R is attached to a rod of same mass and length $2R$ as shown in the figure. A horizontal force F is applied at the top. Sufficient friction is present between the ring and ground to prevent slipping. The acceleration of c.m. is

(A) $\frac{6F}{7m}$ (B) $\frac{3F}{5m}$ (C) $\frac{2F}{3m}$ (D) $\frac{F}{2m}$



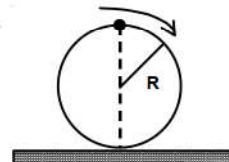
- Q.16 A horizontal force F is applied at the top of an equilateral triangular block having mass m and side a as shown in figure. The coefficient of friction required to topple the block before translation will be

(A) 0.45 (B) 0.35 (C) 0.65 (D) 0.54



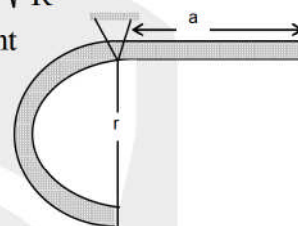
- Q.17 A particle of mass m is attached to a ring of equal mass. The system is released from rest as shown in the figure. Assuming rolling motion the angular velocity of the ring after it turns through 180° is

(A) $\sqrt{\frac{2g}{R}}$ (B) $\sqrt{\frac{g}{R}}$ (C) $\sqrt{\frac{g}{2R}}$ (D) $\sqrt{\frac{3g}{R}}$



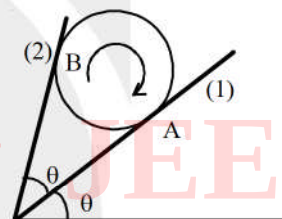
- Q.18 A wire is bent in form as shown in the figure. It is suspended from the point from where curved portion starts (curved portion is semicircular). Find value of a in terms of r such that straight portion remains horizontal.

(A) $2r$ (B) $4r$ (C) $\frac{3}{2}r$ (D) none of these



- Q.19 A sphere of mass m is rotating between two inclined walls with a uniform angular velocity as shown in the figure. If the coefficient of friction between each wall and the sphere is μ . Then the ratio of the friction forces (f_1/f_2) offered by the walls 1 and 2 on the sphere will be

(A) $\tan \theta$ (B) $\tan 2\theta$ (C) $\frac{\sin 2\theta - \mu \cos 2\theta}{\sin \theta - \mu \cos \theta}$ (D) $\cos 2\theta$



[MULTIPLE CORRECT CHOICE TYPE]

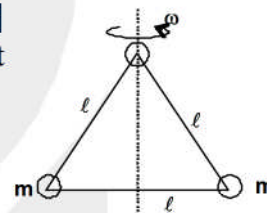
- Q.20 Three identical particles each of mass m are attached by massless rods kept along the sides of an equilateral triangle of side l . If the system rotates with a constant angular speed ω about a vertical axis,

(A) The energy of rotation is equal to $\frac{m l^2 \omega^2}{4}$

(B) The tension in horizontal rod is zero.

(C) The tension in the inclined rod is $\frac{2mg}{\sqrt{3}}$

(D) The velocities of the masses at the end of the horizontal rod are equal.



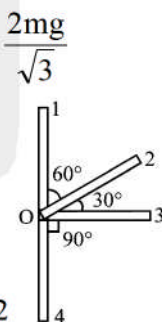
- Q.21 A thin rod free to rotate about a horizontal axis through one end O is initially in unstable equilibrium position 1 and falls due to gravity on being displaced infinitesimally. In positions 2, 3 and 4 it makes 60° , 90° and 180° with the upward vertical. Let ω_2 , ω_3 , and ω_4 be angular velocities of the rod in these positions then

(A) $\omega_4 = 2\omega_3$

(B) $\omega_4 = 2\omega_2$

(C) $\omega_3 = 1.5\omega_2$

(D) $\omega_3 = \omega_2\sqrt{2}$



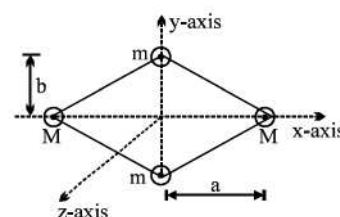
- Q.22 Four point masses are fastened to the corners of a frame of negligible mass lying in the xy plane. Let ω be the angular speed of rotation. Then

(A) rotational kinetic energy associated with a given angular speed depends on the axis of rotation.

(B) rotational kinetic energy about y -axis is independent of m and its value is $Ma^2\omega^2$.

(C) rotational kinetic energy about z -axis depends on m and its value is $(Ma^2 + mb^2)\omega^2$.

(D) rotational kinetic energy about z -axis is independent of m and its value is $Mb^2\omega^2$.



Q.23 A thin hemispherical shell of uniform density and radius R is placed upside down on a rough inclined plane. As the angle of the inclined plane is gradually increased, the shell starts toppling without slipping. Which of the following is true?

(A) The angle of inclined plane with horizontal must be $\tan^{-1}(2)$.

(B) Centre of mass of the shell should be vertically above the lowermost point of the shell.

(C) $\mu > 2$

(D) $\mu < \frac{3}{2}$

[MATRIX TYPE]

Q.24 Column-I gives some direction of the force exerted by surface on the body. Match them with situations in column-II.

Column-I

(A) Direction of force can be in Ist quadrant.

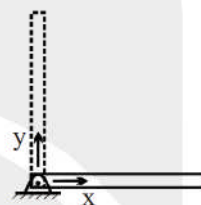
(B) Direction of force can be in IInd quadrant.

(C) Direction of force can be in IIIrd quadrant.

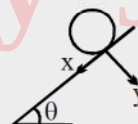
(D) Direction of force can be in IVth quadrant.

Column-II

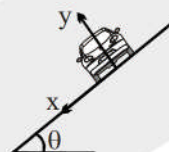
(P) A rod hinged at end is released from vertical position. When it reaches horizontal position, force by the hinge.



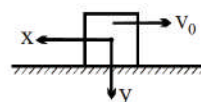
(Q) A sphere rolling down the incline. The force by the inclined surface on the sphere



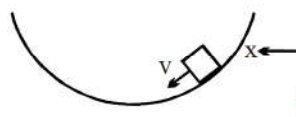
(R) A car moving on a banked circular track with a uniform speed greater than $\sqrt{Rg \tan \theta}$. Force exerted by the track on the car.



(S) A block sliding on a rough level ground force exerted by the ground on block.

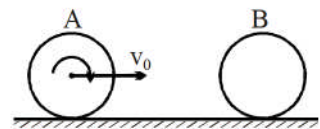


(T) A block moving on a rough circular path. Force exerted by the path.



[SUBJECTIVE TYPE]

- Q.25 A hollow spherical ball A rolling ($v_0 = 10 \text{ m/s}$) without slipping on rough horizontal plane strikes another identical ball head on as shown in the figure. Collision is perfectly elastic. During the collision, the angular velocity of either of the balls does not change. Velocity (in m/s) of ball B when it starts pure rolling is



ANSWER KEY

Q.1	C	Q.2	D	Q.3	B	Q.4	A	Q.5	B	Q.6	B	Q.7	D
Q.8	A	Q.9	C	Q.10	B	Q.11	C	Q.12	B	Q.13	A	Q.14	B
Q.15	B	Q.16	C	Q.17	A	Q.18	D	Q.19	C	Q.20	A,C	Q.21	B, D
Q.22	A,B,C			Q.23	A,B,C			Q.24	(A) R (B) P (C) QT (D) ST				
Q.25	6 m/s												

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